MKE | Acuan cepat satu halaman

Ringkasan rumus untuk disalin ke catatan; lihat versi belajar untuk asumsi, diagram, dan penjelasan.

PISTON & BAHAN BAKAR

$V_{d,total} = z(\pi/4)D^2L$; $r = (V_d+V_c)/V_c$.
$a = 2$ (4 langkah), $a = 1$ (2 langkah).
$P_i = IMEP\ V_{d,total}\ N/(60a)$.
$P_\beta = 2\pi N\tau/60$; $P^f = P_i - P_\beta$.
$\eta_m = P_\beta/P_i$; $BMEP = \eta_m\ IMEP$.
$\eta_{\beta,th} = P_\beta/(\dot{m}f\ LHV)$; $\eta_{\beta,th} = \eta_m\eta_{i,th}$.
$\dot{m}f = P_\beta/(\eta_{\beta,th}\ LHV)$.
$BSFC[g/kWh] = 3{,}6\times10^6/(\eta_{\beta,th}\ LHV[kJ/kg])$.
$\eta comb \neq \eta_{i,th}$; jangan samakan keduanya.
Daya MEP: kPa,m³ → kW; torsi: N·m → W.

DASAR & SATUAN

$Q - W = \Delta U$; gas ideal $pv = RT$.
$\Delta u = c_v\Delta T$; $\Delta h = c_p\Delta T$; $k = c_p/c_v$.
$R = c_p - c_v$; isentropik: $pv^k = C$.
$T_2/T_1 = (p_2/p_1)^{((k-1)/k)}$.
Isobarik: $w = R\Delta T$; $q = c_p\Delta T$.
Isokhorik: $w = 0$; $q = c_v\Delta T$.
Adiabatik tertutup: $w = c_v(T_{in} - T_{out})$.
1 bar = 100 kPa; 1 cc = 10^{-6} m³.
1 kWh = 3600 kJ; 1 ton/h = 1000/3600 kg/s.
1 hp ≈ 0,7457 kW; T[K] = T[°C]+273,15.

OTTO & DIESEL — k KONSTAN

Otto: s=konstan → v=konstan →
s=konstan → v=konstan.
$T_2 = T_1 r^{(k-1)}$; $T_4 = T_3/r^{(k-1)}$.
$q_{in} = c_v(T_3 - T_2)$; $q_{out} = c_v(T_4 - T_1)$.
$\eta Otto = 1 - 1/r^{(k-1)}$.
Diesel: s=konstan → p=konstan →
s=konstan → v=konstan; $\rho = v_3/v_2$.
$T_3 = \rho T_2$; $T_4 = T_3(\rho/r)^{(k-1)}$.
$q_{in} = c_p(T_3 - T_2)$; $q_{out} = c_v(T_4 - T_1)$.
$\eta Diesel = 1 - (\rho^k - 1)/[k(\rho-1)r^{(k-1)}]$.
$w_{net} = q_{in} - q_{out}$; $\eta = w_{net}/q_{in}$.

BRAYTON & BOILER

1→2 kompresor; 2→3 bakar; 3→4 turbin.
$w_c = h_2 - h_1$; $w_t = h_3 - h_4$.
$q_{in} = h_3 - h_2$; $w_{net} = w_t - w_c$.
$BWR = w_c/w_t$; $P_{net} = \dot{m}\ w_{net}$.
$\dot{m}_{air} = p\dot{V}/(RT)$; $\eta = w_{net}/q_{in}$.
$\eta Brayton\ ideal = 1 - 1/r_p^{((k-1)/k)}$.
Boiler: $\dot{Q} = \dot{m}s(h_{out} - h_{in})$.
$\dot{m}f = \dot{Q}/(\eta boiler\ LHV)$.
$HR = 3600/\eta fuel\to elec\ [kJ/kWh]$.
$\eta fuel\to elec = \eta boiler\ \eta cycle\ \eta gen$
hanya jika $\eta cycle$ berbasis kalor fluida.

RANKINE & TABEL UAP

1 kondensor keluar; 2 pompa keluar;
3 boiler keluar; 4 turbin keluar.
$w_p = h_2 - h_1 \approx v_1(p_2 - p_1)$.
$w_t = h_3 - h_4$; $q_{in} = h_3 - h_2$.
$\eta = (w_t - w_p)/q_{in}$; $P_{net} = \dot{m}(w_t - w_p)$.
$\eta_t = (h_{in} - h_{out,a})/(h_{in} - h_{out,s})$.
$\eta_p = (h_{out,s} - h_{in})/(h_{out,a} - h_{in})$.
$x = (s - s_f)/s_{fg}$; $h = h_f + xh_{fg}$.
Jenuh kering: $x=1$; cair jenuh: $x=0$.
Perbaikan: superheat, pcond turun,
pboiler naik, reheat, regenerasi.

TEORI & PEMERIKSAAN

ICE: pembakaran internal; piston/turbin gas.
ECE: kalor eksternal; uap/Rankine/Stirling.
HHV/GCV > LHV/NCV: kalor kondensasi
air produk termasuk hanya dalam HHV.
SI knock: autoignition end-gas.
Diesel knock: delay → pembakaran cepat.
Oktan: tahan knock SI; cetane: mudah nyala CI.
η vs rpm aktual bergantung beban.
Isentropik = adiabatik + reversibel.
$P_\beta \leq P_i$; η harus 0–1; $q_{in} > 0$.
Pastikan volume total, basis fuel, dan satuan.

Prioritas dari contoh ujian: performa piston • ICE/ECE • Brayton • boiler/heat rate • Rankine • knocking.